

SECP3744 - UTHM CASE STUDY
UTHM INVENTORY SYSTEM PROJECT REPORT

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ON-LINE RECOGNITION OF DEVELOPING CONTROL CHART PATTERNS
SECP3744 - UTHM CASE STUDY
UTHM INVENTORY SYSTEM PROJECT REPORT

HARINI A/P SANGARAN

A project report submitted in fulfilment of the
requirements for the award of the degree of
Bachelor of Computer Science (Data Engineering) with Honours

School of Education
Faculty of Computing
Universiti Teknologi Malaysia

DECEMBER 2025

DECLARATION

I declare that this project report entitled “*UTHM INVENTORY SYSTEM PROJECT REPORT*” is the result of my own research except as cited in the references. The project report has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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ABSTRACT

This report presents a case study on UTHM focusing on the analysis and improvement of its existing digital systems through the development of a functional Enterprise Architecture (EA). During an industrial visit, it was observed that although UTHM is in the process of establishing a new data centre and has adopted various digital systems, a comprehensive enterprise architecture that clearly represents system integration, data management, and technology infrastructure is still lacking due to limited expertise. The objective of this study is to analyse the current system environment and propose an improved enterprise architecture that aligns with UTHM's digital transformation goals. As part of the proposed solution, a SAP-based inventory management system was designed to support stock checking, inventory updates, and report generation, demonstrating how enterprise systems can be integrated effectively within the proposed architecture. This report covers the current system analysis, proposed enterprise architecture, system design diagrams, and interface mock-ups to illustrate the feasibility of the solution. The findings highlight how a structured enterprise architecture and the adoption of enterprise systems such as SAP can enhance operational efficiency, data consistency, and system governance at UTHM.

ABSTRAK

Laporan ini membentangkan kajian kes mengenai UTHM yang memberi tumpuan kepada analisis dan penambahbaikan sistem digital sedia ada melalui pembangunan Seni Bina Perusahaan (EA) yang berfungsi. Semasa lawatan industri dijalankan, didapati bahawa walaupun UTHM dalam proses membangunkan pusat data baharu dan telah menggunakan pelbagai sistem digital, satu seni bina perusahaan yang komprehensif untuk menggambarkan integrasi sistem, pengurusan data, dan infrastruktur teknologi secara jelas masih berkurangan disebabkan oleh kepakaran yang terhad. Objektif kajian ini adalah untuk menganalisis persekitaran sistem semasa dan mencadangkan penambahbaikan seni bina perusahaan yang selaras dengan matlamat transformasi digital UTHM. Sebagai sebahagian daripada penyelesaian yang dicadangkan, sebuah sistem pengurusan inventori berasaskan SAP telah direka untuk menyokong semakan stok, pengemaskinian inventori, dan penghasilan laporan, sekali gus menunjukkan bagaimana sistem perusahaan boleh disepadukan secara berkesan dalam seni bina yang dicadangkan. Laporan ini merangkumi analisis sistem semasa, cadangan seni bina perusahaan, gambar rajah reka bentuk sistem, dan rekaan antara muka (mock-ups) bagi menggambarkan kebolehlaksanaan penyelesaian tersebut. Hasil kajian menekankan bagaimana seni bina perusahaan yang terstruktur dan penggunaan sistem perusahaan seperti SAP dapat meningkatkan kecekapan operasi, konsistensi data, dan tadbir urus sistem di UTHM.

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LIST OF ABBREVIATIONS

UTHM	-	Universiti Tun Hussein Onn Malaysia
EA	-	Enterprise Architecture
TOGAF	-	The Open Group Architecture Framework (TOGAF)
IT	-	Information Technology
SA	-	System Architecture
ERD	-	Entity Relationship Diagrams
LMS	-	Learning Management System
UiTM	-	University Teknologi MARA
SAP	-	Systems, Applications, and Products in Data Processing
BTP	-	Business Technology Platform
SDLC	-	System Development Life Cycle
SSO	-	Single Sign-On
WAF	-	Web Application Firewall
API	-	Application Programming Interface
LAN	-	Local Area Network
VPN	-	Virtual Private Network

CHAPTER 1

INTRODUCTION

1.1 Overview

Universiti Tun Hussein Onn Malaysia (UTHM) is in the process of constructing a new on-premise data centre to facilitate the consolidation, modernisation, and oversight of its institutional Information Technology (IT) infrastructure. Notwithstanding this strategic commitment, the institution does not own a formalised Enterprise Architecture (EA) that can deliver a cohesive framework aligning business processes, information systems, data assets, and foundational technical platforms. The lack of a formalised Enterprise Architecture has resulted in disjointed system implementations, segregated data repositories, restricted visibility of inter-system relationships, and considerable difficulties in achieving interoperability and effective data integration across many departments and functional areas. These conditions impair operational efficiency, restrict data-driven decision-making, and obstruct UTHM's capacity to attain sustainable digital transformation. In acknowledgement of these limitations, UTHM has demonstrated a willingness to accept external and academic contributions that can present a complete and adaptable EA model appropriate for institutional application.

This thesis addresses the institutional requirement by developing a comprehensive Enterprise Architecture for UTHM, informed by The Open Group Architecture Framework (TOGAF). TOGAF is utilised for its globally acknowledged structured methodology, focus on governance, and appropriateness for large, complex entities like higher education institutions. The proposed EA aims to create a cohesive and standardised architectural framework that aligns UTHM's strategic goals with its technological capacities, enhances integration among academic and administrative systems, improves governance and lifecycle management of IT assets, and facilitates scalable future digital initiatives. The study further creates a prototype enterprise system utilising SAP technology to operationalise and test the suggested design. This

prototype illustrates the actual application of essential architectural elements, highlights the integration of enterprise resource processes within the EA framework, and assesses the viability of SAP as a fundamental platform to underpin UTHM's mission-critical operations.

This research seeks to furnish UTHM with a systematic, pragmatic, and future-oriented Enterprise Architecture model by integrating architectural design and system prototyping, thereby reinforcing institutional governance, fostering system interoperability, improving data integration, and advancing the university's enduring digital modernisation strategy within its new data centre environment.

1.2 Problem Statement

UTHM is in the process of establishing a new on-premise data center to support the centralization and modernization of its institutional IT infrastructure. UTHM currently does not possess a functional EA that clearly represents how its business processes, applications, data, and technology components are integrated and governed. This is because UTHM lacks EA expertise.

The lack of a well-defined EA has resulted in fragmented system management, limited visibility of system interdependencies, and difficulties in ensuring effective data integration. In order to solve this problem, UTHM has stated that it is open to external proposals, including academic contributions, that could demonstrate a suitable enterprise architecture model for potential adoption.

In response to this challenge, this case study focuses on improving the current system landscape in order to design an enhanced and functional Enterprise Architecture for UTHM. In addition, a prototype system utilizing SAP technologies is developed to allow UTHM to test an integrated enterprise system within the proposed architecture. This approach aims to demonstrate how SAP can be aligned with the EA to support core university operations, improve data integration, and validate the practicality of the proposed architecture within UTHM's new data center environment.

1.3 Project Objectives

- To design a comprehensive architectural framework that aligns the inventory system with UTHM's organizational structure and ICT policies.
- To model and automate business processes that allow seamless handovers between departments.
- To establish a Single Source of Truth by designing an integrated enterprise data model and eliminating data redundancy and ensuring consistent asset information across all UTHM divisions.
- To develop a system capable of generating enterprise-level insights, helping UTHM management make data-driven decisions on budget planning and asset utilization.

1.4 Project Scope

The scope of this project is designed to fit in the project where this project focuses on providing a structured transition from UTHM's existing enterprise model to an integrated enterprise model based on their goals in digitalizing transformation. The main components of the project are:

1. Review existing system enterprise model

This project begins with an "As-Is" assessment of UTHM's current system environment based on the visit institutional observations. This phase focuses on analyzing how redundant processes and the lack of real-time data sharing hinder operational efficiency. The goal is to document the current relationship between business processes and evaluate how the current system lack of integration impacts institutional operational efficiency.

2. Design To-Be Enterprise Model

The project involves designing a comprehensive "To-Be" Enterprise Model that provides a scalable and integrated blueprint for the university that maps the four architectural layers (Business, Data, Application, and Technology) The scope includes a detailed design of Use Case Diagrams, Use Case Descriptions, Activity

Diagrams, and Sequence Diagrams to model the behavioral logic of the subsystems. Additionally, it covers Database Design through Class Diagrams and Entity Relationship Diagrams (ERD) to ensure data integrity, and designing the interface to define the user experience and interaction for the individual subsystem within the new architecture.

3. System Development

This project also involves building a working prototype using the SAP Business Technology Platform to connect different departments and show how information flows smoothly. In this stage, the physical database is created to store all necessary data, and the main system functions are coded. This process demonstrates how an automated, integrated system makes university operations faster and more efficient.

1.5 Project Importance

This project is important as it helps UTHM for addressing the current problems especially in UTHM's enterprise architecture at the same time to improve its operational efficiency, system integration and user experience. The project ensures that data flows smoothly across platforms like Learning Management System (LMS) and student portal by analyzing and enhancing the integration of subsystems such as student enrollment, course registration, hostel management and university healthcare. This improvement surely benefits students, staff and administrators by providing them with reliable access to information and services.

Optimising proposal enterprise architecture helps UTHM make better decisions, use resources more effectively, and ensure their IT systems can grow over time. This approach will take full advantage of new technologies like AI, IoT, and cloud computing to support their academic, administrative, and research work. In the end, this project strengthens UTHM's commitment to digital transformation, operational excellence, and ongoing improvement, helping them remain a modern and efficient university.

CHAPTER 2

LITERATURE REVIEW

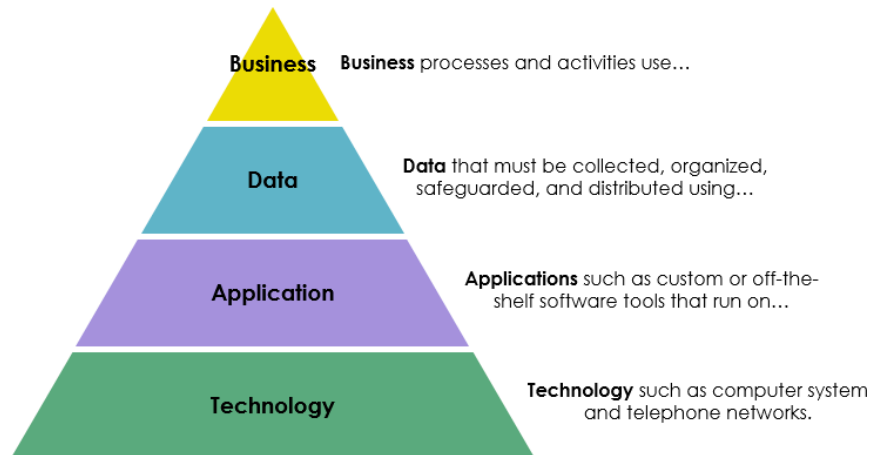
2.1 Introduction

This chapter provides a literature review of the key concepts that define the enterprise-level design and integration discussed in this project. The modern enterprise, especially in higher education, requires a comprehensive framework to strategically align its business objectives with its IT capabilities and infrastructure. This framework is known as EA. We will also discuss the related sub-system that exists in universities EA by taking Universiti Teknologi MARA (UiTM) as an example. In the end, we are going to discuss technology used in EA which is the SAP Business Technology Platform (BTP) and its details.

2.2 Enterprise Architecture (EA)

EA is defined as a comprehensive framework designed to manage and align an organization's business objectives with its IT capabilities and infrastructure (BOC Group, 2023; CIO Index, n.d.). It encompasses the systematic design, planning, and management of the organization's technology, processes, and systems to effectively support its strategic goals and objectives (BOC Group, 2023). EA provides a holistic view of the enterprise by organizing its structure into four interconnected architectural domains (Visual Paradigm, 2024)

Figure 2.1: *The Four Domains of Enterprise Architecture*



Note. Source: (Visual Paradigm, 2024).

Business Architecture: This domain maps the organization's strategy, governance, core business capabilities, value streams, and processes. Its central focus is on defining the "what" and "why" of the business operations, ensuring that technological investments are driven by strategic needs (CIO Index, n.d.).

Data Architecture: This domain establishes the principles and structures for how the organization's data assets are to be collected, stored, managed, and utilized. It is critical for ensuring data quality, consistency, and accessibility throughout the entire enterprise (BOC Group, 2023).

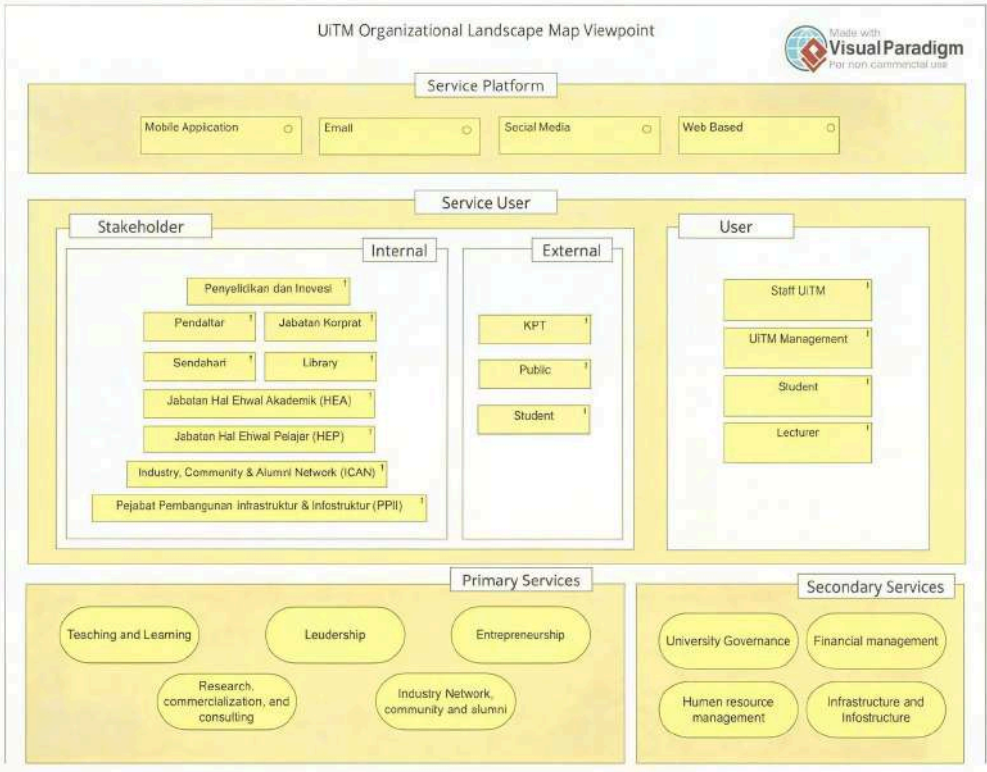
Application Architecture: This domain involves identifying the specific applications utilized within the enterprise and defining their interactions with each other and with external stakeholders. The objective is to establish an integrated, efficient, and scalable application portfolio (CIO Index, n.d.).

Technology Architecture: This domain describes the foundational hardware, software, networking infrastructure, and operating systems that collectively support the application and data layers. Its primary concern is ensuring the reliability, scalability, and security of the underlying technology foundation (BOC Group, 2023; CIO Index, n.d.).

2.3 Related Subsystem

EA in higher education institutions typically integrates multiple subsystems that support academic, research, governance, administration, and community functions. These subsystems interact across different layers, including business processes, service delivery, information technology platforms, and stakeholder interactions, in an attempt to align institutions with their strategic goals. Many universities adopt EA for simplifying services provided to students and staff, standardising information management, and increasing decision-making effectiveness (Amin et al., 2024).

Figure 2.2: *UiTM Organizational Landscape Map Viewpoint*



Note: Source: (Amin et al., 2024).

Figure 2.2 shows UiTM’s EA landscape, which is built around different subsystems. Each subsystem handles specific tasks, such as academic services, administrative services and digital platforms. They all work together to connect stakeholders and service users. The framework includes internal and external stakeholders, main and supporting service areas and works across platforms like mobile, email, social media, and websites. By linking these subsystems, UiTM creates a smooth and integrated

service environment that supports its goals for improving the institution (Amin et al., 2024).

Table 2.1 *Description of UiTM EA Subsystems*

Subsystem Category	Subsystem	Description
Service Platforms	Mobile App, Email, Social Media, Web-based Systems	Digital channels enabling communication and service access for users.
Stakeholders	Internal: HEA, HEP, Library, Corporate, ICAN, PPII, etc.	Organizational units involved in academic, student, infrastructure, alumni, and corporate services.
	External: KPT, Public, External Students	External bodies influencing or consuming UiTM services.
User Groups	UiTM Staff, Management, Students, Lecturers	Primary service users interacting with UiTM digital and administrative services.
Primary Services	Teaching & Learning, Leadership, Entrepreneurship, Research & Commercialization, Alumni & Industry Network	Core mission-driven services supporting UiTM’s academic and developmental goals.
Secondary Services	University Governance, Human Resource, Finance, Infrastructure	Supporting subsystems enabling operations, compliance, and resource management.

Table 2.1 provides an overview of how each subsystem at UiTM supports the services offered at the university and the involvement of different stakeholders. It categorises subsystems into functional types, describes their purpose, and identifies service platforms on which users can access them. It also provides information on the internal and external stakeholders and user groups, and the primary and secondary services each subsystem enables. This structured view shows how UiTM integrates its digital and administrative systems into a seamless, mission-driven service environment.

Table 2.2 *Comparison between Layered EA Model and UiTM Organizational Landscape Map Viewpoint*

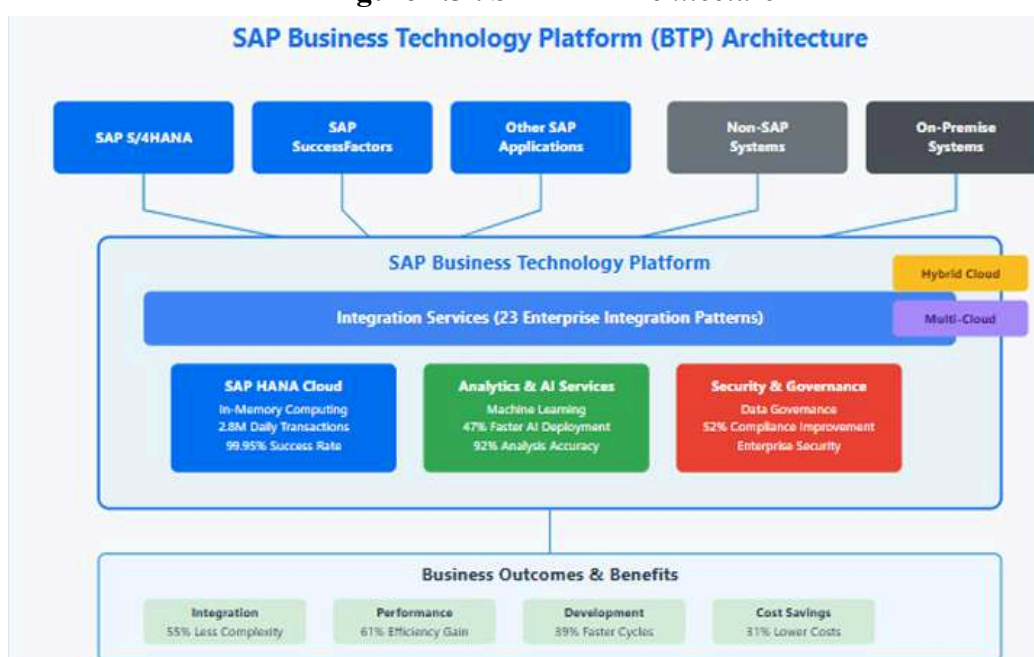
Feature	Layered EA Model	UiTM Organizational Landscape Map Viewpoint
Main Focus	Architectural Components (What are the components of the enterprise's IT?).	Organizational and Service Components (Who uses the services and what services are offered?).
Business/Organizational Content	Lists Functional Areas : Academic Management, Finance, Human Resources, etc.	Lists Key Stakeholders (Internal/External), Users (Staff, Student, Lecturer), and Services (Teaching, Leadership, Governance).
IT/Technology Content	Lists Abstract Domains : Servers, Network, Middleware, Database (under Technology Architecture).	Lists Delivery Channels/Platforms : Mobile Application, Email, Social Media, Web Based (under Service Platform).
Framework Link	Strongly aligned with the TOGAF/Zachman frameworks' core layered structure.	Represents a Viewpoint or business-centric model, often part of an overall EA deliverable.

This Table 2.2 visual two representations of EA serve distinct but complementary roles: the Layered EA Model acts as a structural blueprint, primarily focusing on Architectural Components by detailing the required Functional Areas like Academic Management and the underlying abstract IT Domains such as Servers and Network, consistent with the formal, tiered structure of frameworks like TOGAF; conversely, the UiTM Organizational Landscape Map Viewpoint functions as a high-level, business-centric communication tool, focusing on Organizational and Service Components by illustrating the key Stakeholders (Staff, Students) and the Delivery Channels (Mobile Apps, Web) they use to access the university's services.

2.4 Technology Used

The SAP BTP is architecturally designed as a unified cloud foundation, acting as a crucial integration layer for the modern enterprise (Komarina & Sajja, 2025). It is built upon a framework that encompasses the Integration Suite, Extension Suite, and Data Management Suite, leveraging a microservices approach to provide a platform with defined scalability features and a structured security framework (Annanki, 2023).

Figure 2.3 : SAP BTP Architecture



Note: Source: (Komarina & Saja, 2025).

Core Components of SAP BTP

1. Database and Data Management

Database and data management focuses on efficient data storage, processing, and governance. Key services include SAP HANA Cloud, SAP Data Warehouse Cloud, and SAP Data Intelligence, which collectively support real-time data access and enterprise-level data management.

2. Analytics

SAP BTP includes analytical tools such as SAP Analytics Cloud (SAC),

enabling organizations to perform reporting, data visualization, predictive analytics, and planning. These tools help businesses make informed, data-driven decisions.

3. Application Development and Automation

This component offers capabilities for building and extending applications using low-code/no-code tools, cloud-native development frameworks, and process automation. Tools such as SAP Build Apps, SAP Build Process Automation, and the SAP Cloud Application Programming Model (CAP) streamline application development and workflow automation.

4. Integration

SAP BTP supports seamless connectivity between SAP and non-SAP systems through services like SAP Integration Suite, API Management, and Event Mesh. These tools ensure smooth data exchange and interoperability across the organization's technology landscape.

SAP BTP enables organizations to modernize business processes, reduce dependency on custom code in core systems, and create flexible, scalable solutions. By unifying application development, data management, analytics, and integration services, the platform helps businesses enhance innovation, improve efficiency, and adapt quickly to changing operational needs.

2.5 Summary

This chapter establishes the theoretical foundation for the project by exploring EA as a tool to align university business goals with IT infrastructure. By examining the four key domains which are Business, Data, Application, and Technology, the review clarifies how a system must be structured to support an entire organization.

Through the case study of UiTM's EA landscape, the chapter illustrates how various subsystems and stakeholders interact to create an integrated service environment. Finally, the review of SAP BTP identifies the modern technology

needed to achieve this integration, focusing on data management, analytics, and seamless system connectivity. Together, these elements provide the blueprint for designing the UTHM Inventory System as a robust enterprise solution.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter is a description of the methodology of Waterfall with System Development Life Cycle (SDLC) used to help with the planning, analysis, design and implementation of improvements to UTHM's Enterprise Architecture framework. The Waterfall of SDLC uses formal, creating or altering systems, and the models and methodologies that people use to develop these systems (Kute & Thorat, 2014). The choice to use a structured methodology focused on requirements, architectures and specifications being fully defined and validated earlier in the process was due to the increased difficulty in changing or correcting Enterprise Architecture projects after they have been implemented due to their high cost of implementation and complexity to implement from start to finish. The following chapters explain each phase of the methodology, including phases, activities and deliverables, provide a project schedule like for example using a Gantt Chart or Kanban format with major milestones, and summarise the entire development of the project.

3.2 Proposed Methodology

This study used the Waterfall SDLC model as its approach. The Waterfall methodology is chosen for its systematic, documentation-centric, and linear characteristics, making it appropriate for major business projects like the creation of EA for UTHM. Considering that EA projects entail substantial implementation costs, widespread institutional effects, and long-term strategic coherence, a methodology prioritising meticulous planning, requirement verification, architectural delineation, and regulated execution is necessary. The Waterfall SDLC facilitates thorough analysis and robust architectural design prior to deployment, reducing rework and maintaining architectural consistency throughout the project lifecycle.

This study uses the Waterfall technique to direct the design and development of an improved Enterprise Architecture framework for UTHM, alongside the creation of an enterprise-wide SAP prototype to validate the proposed architecture. Each phase of the technique concentrates on systematically converting UTHM's institutional requirements into a structured design, subsequently validated through technical implementation. Stakeholder interaction is integrated to guarantee conformity with institutional expectations and operational reality. The suggested methodology has five primary phases: Feasibility Study, System Analysis, System Design, Implementation, and Maintenance and Support.

3.3 Phases of Methodology

The project follows the five core phases of the SDLC which is Feasibility Study, System Analysis, System Design, Implementation and Maintenance and support

Phase 1: Feasibility Study

This first stage will determine whether the project is viable. We adhere to UTHM's standard operating procedure for new system requests.

- FSR Process: The project began with the submission of the Feasibility Study Report (FSR).
- Committee Approval: FSR is submitted to the IT Security & Steering Committee which will evaluate the proposal.

Outcome: The "Go /No-Go" decision based on the project proposal and literature review.

Phase 2: System Analysis

In this phase , we collect the detailed business requirements to bridge the gap between the definition of the requirement and how it should be.

- Data Collection: Conducting interviews with UTHM stakeholders (Asset Unit officers) to understand the current manual form and workflow.

- Requirement Specification: Defining the user roles and their access privileges.

Phase 3: System Design

In this phase , we convert the requirements into technical specifications.

- Architecture Design: We utilize the three-tier architecture model to ensure the system is scalable and easy to maintain.
- Database Design: Create the entity-relationship diagram to model the relationships between asset, staff and location.
- Interface Design: Prototyping the dashboard UI to ensure they are user-friendly for non technical staff.
- Infrastructure Design: The system will be hosted in UTHM's Data Center. It will benefit from high-availability infrastructure that includes dual power sources, UPS backup and generator set to ensure very high uptime.

Phase 4: Implementation

This is the phase where the system is built and implemented.

- Coding: Developing the subsystem using web technologies such as PHP and java then connect it to the database MySQL.
- Data Migration: Adopt a phased migration strategy, which converts sample data from existing Excel sheets into the new system format.
- Security Implementation: The code will be hosted behind the UTHM's existing security layer which include 2 Firewalls (Europe/China) and a Web Application Firewall to protect the portal from being attacked from the external side.
- Testing: Perform testing on each individual module and integration testing to ensure it communicates with the Financial module.

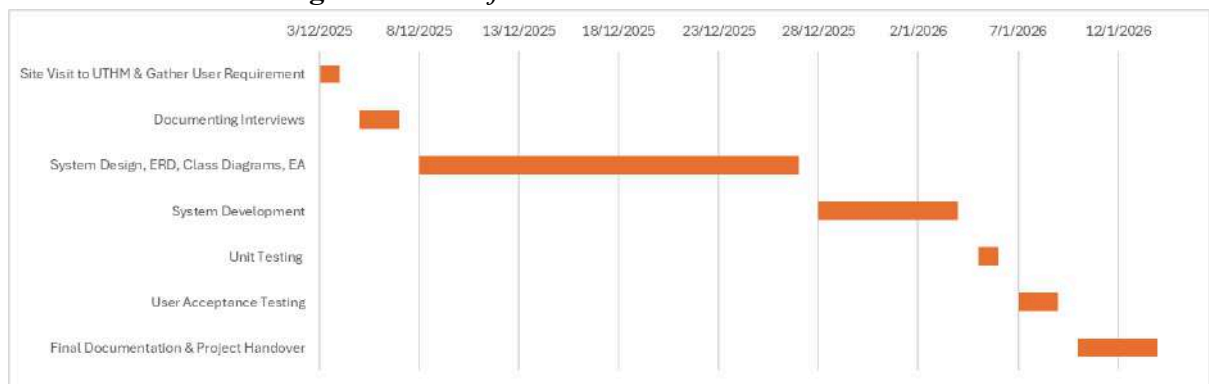
Phase 5: Maintenance and Support

In this phase , we involve training and patching activities. We will teach the UTHM staff how to use the new QR code features and fix any bugs that are discovered after the system is deployed.

3.4 Project Planning Schedule

In order to make sure this project runs smoothly, a schedule consists of every phase and its duration is made. The team members that are in charge of leading the task are also determined to make sure all the phases are supervised.

Figure 3.1 : Project Schedule Gantt Chart



Tasks	Start Date	End Date	Task Leader
Site Visit to UTHM & Gather User Requirement	3/12/2025	4/12/2025	Aisyah
Documenting Interviews	5/12/2025	7/12/2025	Harini
System Design, ERD, Class Diagrams, EA	8/12/2025	27/12/2025	Zhiming
System Development	28/12/2025	4/1/2026	Afiq
Unit Testing	5/1/2026	6/1/2026	Auni
User Acceptance Testing	7/1/2026	9/1/2026	Ziyaad
Final Documentation & Project Handover	10/1/2026	14/1/2026	Ika

3.5 Summary

This chapter has detailed the Waterfall SDLC methodology chosen to guide the improvement of UTHM's Enterprise Architecture. By utilizing the five core phases which are feasibility study, system analysis, system design, implementation, and maintenance, the project minimizes the risks associated with the high cost and complexity of enterprise-level systems.

This structured path ensures that the resulting SAP prototype is not only technically sound but also strategically aligned with UTHM's specific operational needs and secure within its new data center environment. The inclusion of a detailed project schedule and Gantt chart ensures that all tasks, from the initial site visit to system implementation and testing are supervised and completed within the established timeframe. Ultimately, this structured methodology serves as the bridge between UTHM's strategic digital transformation goals and the successful delivery of a secure, integrated, and high-performance enterprise environment.

CHAPTER 4

Analysis and Design

4.1 Introduction

This chapter analyses the organisational structure of UTHM, the current system environment, and enterprise requirements to facilitate the development of a complete EA framework and a SAP-based enterprise prototype. A comprehensive comprehension of the university's organisational structure, operational processes, and information dissemination methods is crucial for ensuring that the suggested design is congruent with institutional goals and effectively tackles existing operational issues. Given that UTHM presently functions with several autonomous systems exhibiting minimal integration, this analysis establishes the essential groundwork for delineating how a cohesive and centralised enterprise framework might improve institutional efficiency, governance, and data-informed decision-making.

The chapter initially analyses UTHM's organisational framework at the Strategic, Tactical, and Operational tiers to pinpoint principal decision-makers, stakeholders, and primary data generators. This is succeeded by a comprehensive evaluation of the existing system landscape, emphasising the dependence on disjointed digital platforms and semi-manual procedures, although the presence of a contemporary three-tier data centre architecture. A comparative analysis of the current and proposed environments is subsequently provided to demonstrate the shift from isolated systems to a fully integrated enterprise ecosystem, facilitated by centralised data management, Single Sign-On (SSO), and an improved user experience.

To guarantee that the suggested architecture appropriately represents institutional demands, requirements gathering activities were performed through stakeholder involvement via structured interviews. These interactions facilitated the recognition of essential business requirements, operational obstacles, technical anticipations, and governance factors. The chapter finishes by delineating the functional and

non-functional requirements that will inform the design, development, and deployment of the improved EA framework. These investigations collectively establish the analytical foundation for the ensuing architectural design and solution implementation phases.

4.2 Company Organization Structure

A successful system integration requires clear understanding of the organizational hierarchy and the information flow between stakeholders. UTHM operates on three distinct levels which are Strategic, Tactical and Operational. This structure must be bridged to eliminate the "Vertical Silos" which are often found in large organizations.

1. Strategic Level

- IT Security and Steering Committee: This high-level body responsible for the overall digital strategy for UTHM. They will review and approve the Feasibility Study Reports for all proposed systems. They need to determine whether the new system is necessary and aligns with the university budget and culture.
- Management Implications: Top management support is critical to enforce the cultural changes to adapt the new system.

2. Tactical Level

- Bursary Department (Bendahari) : Bursary defines the rules that the software must obey. They act like a central hub for data verification, requiring inputs from various faculty to ensure accuracy.

3. Operational Level

- Faculties and Responsibility Centers: This level includes academic and administrative staff that are distributed across the campus. They are the primary data generators.

4.3 Current System Analysis

UTHM relies on a range of digital systems and technologies to support its academic, administrative, and operational activities. Online platforms are used for teaching and

learning, student administration, finance, human resource management, and communication between students and staff. These systems enable basic information access and daily operations. However, they are implemented independently to meet specific departmental needs. As a result, data sharing across departments are limited and operational efficiency depends on manual processes. In terms of infrastructure, UTHM is in the process of developing a new three-tier data center to host and support its digital systems. While this infrastructure provides a strong physical foundation for digital services, its capabilities have not yet been fully utilized in the current system environment as there are no centralized system management and integrated digital solutions. This situation highlights the need for improved system coordination and structured planning to maximize the benefits of the new data center.

4.4 Comparison between Existing and Proposed System

The transition from UTHM's current digital landscape to the proposed architecture represents a shift from a fragmented system to a unified digital system. Currently, the university relies on independent systems that serve specific departmental needs. While a modern three-tier data center exists, its potential remains not fully utilized because the current systems lack the centralized management and integration needed to utilize such a foundation. The proposed system addresses these limitations by replacing manual workarounds with end-to-end digitalization, guided by a clear "Digital Transformation" vision. Instead of isolated data, the proposed architecture introduces a centralized data warehouse and automated pipelines to ensure information flows seamlessly across the institution. This change moves towards a more agile, hybrid cloud environment. Furthermore, the user experience is improved from using multiple disconnected platforms to a streamlined approach featuring SSO and a unified mobile app.

4.5 System Requirements Gathering Technique

To ensure the proposed UTHM Inventory System aligns with the actual operational needs of the university, the team employed the Interview Method as the primary data collection technique. This qualitative approach allowed the team to explore the

complexities of existing workflows and identify specific pain points across different hierarchical levels. The interview method was selected due to its flexibility and the depth of information it provides. Compared to other methods like questionnaires, interviews allow the team to capture the specific needs of both the administrative staff and the end-users. The interviews were conducted during the site visit on 3/12/2025. The team roles were distributed as follows:

- **Lead Interviewer (Aisyah):** Facilitated the discussion using a prepared set of open-ended questions.
- **Technical Observers (Afiq & Ziyaad):** Focused on identifying technical constraints and integration points with current UTHM legacy systems.
- **Note-Takers (Ika & Harini):** Documented the responses and captured non-verbal cues regarding system frustrations.

Below are the sample of questions interviewed and its answers:

Q: Are there still manual methods or all is digitalized?

A: UTHM is not 100% digital yet. One of the departments that are still running manually are their Health Center. However, admin been enforcing lecturer to use LMS and work on a singular system.

Q: Is there a stand alone database different from the main database?

A: Only LMS has separated database but all the system get data in on single database

Q: Who approves of the new system?

A: The team asks for user requirements first, then the team will develop the system and send it to the committee to decide either to accept or reject the new system.

4.6 System Requirements

Functional Requirements

Functional requirements specify the particular functionality and features that an updated EA framework should offer to UTHM to help streamline primary business

operations in UTHM's IT ecosystem. Functional requirements help to ensure that the updated EA framework provides support for key UTHM processes such as asset management, decision making and compliance.

- User Authentication - A secure account accessible for all UTHM staff and students using SSO.
- Requirements Management - All systems requirements should be captured, traced and validated from the SDLC analysis phase and then linked to the design output.
- Reporting - Generate customizable reports that should be exportable along with Gantt charts and phase deliverables that can be viewed by administrators.
- Integration Support - EA Framework must provide the ability to import and export data that is in existing tools like TOGAF during integration activities.

Non-Functional Requirements

Non-functional requirements specify the quality attributes and performance standards the EA framework must meet, ensuring reliability, efficiency, and usability in UTHM's demanding academic environment.

- Performance: System shall load web pages within 3 seconds for datasets up to 1,000 components.
- Scalability: System must support around 2000 concurrent users.
- Security: Encrypt data especially for password and using firewall like Web Application Firewall (WAF) on the webpage system.
- Usability: Intuitive for new users and accessible for various users.
- Reliability: The system must have 99.5% uptime.

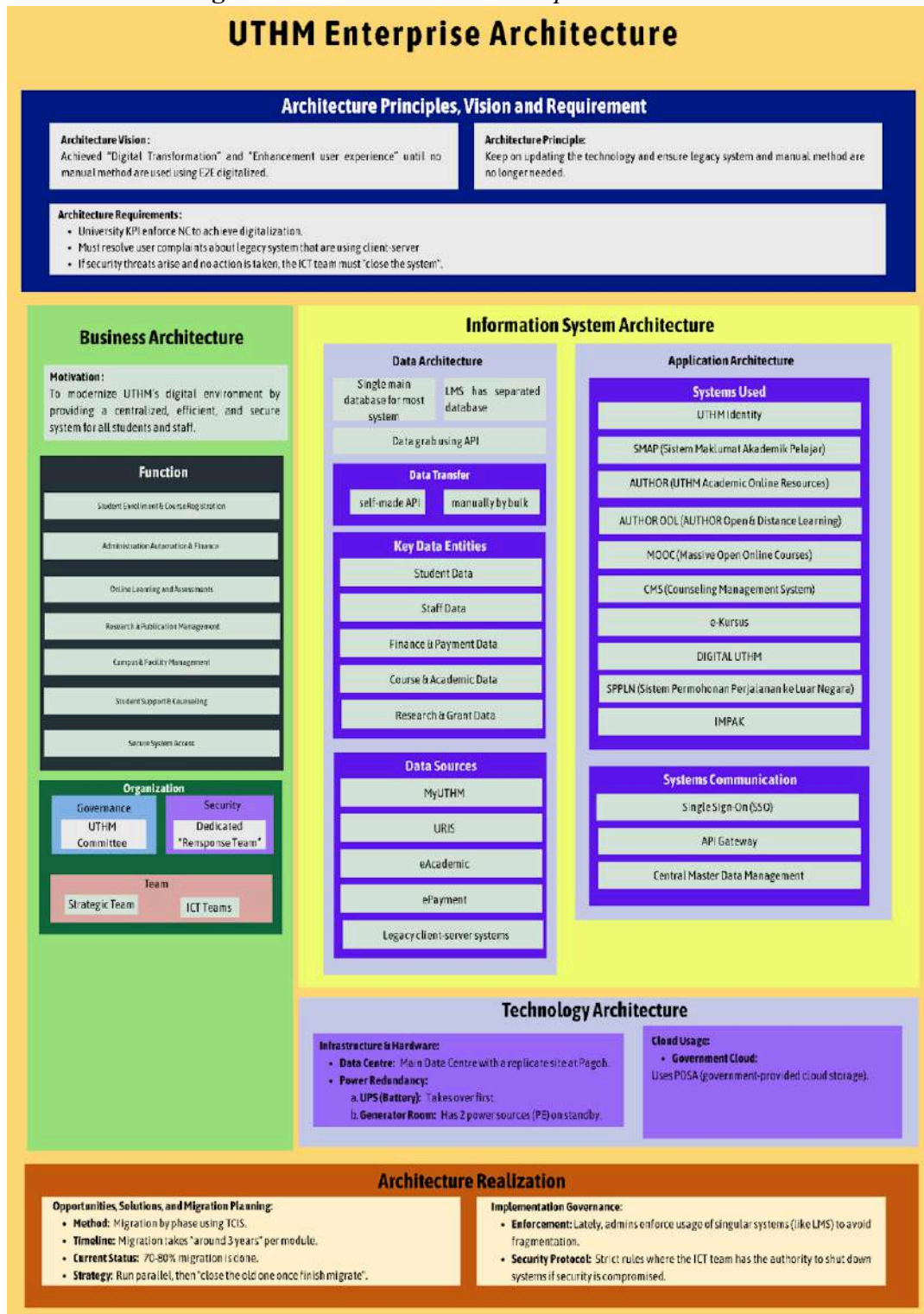
4.7 System Design

This section consists of all of the system designs such as EA As-Is, EA To-Be, System Architecture, Project Design, Database Design and Interface Design.

4.7.1 Enterprise Architecture AS-IS

Figure 4.1 shows the UTHM's As-Is Enterprise Architecture. The detailed explanation for each component is given in 4.7.2.

Figure 4.1 : UTHM's As-Is Enterprise Architecture



4.7.2 System Design AS-IS

This section explains the current architecture at UTHM before the new system is implemented.

1. Business Architecture

- **Strategy:** Guided by “Digital Transformation 2030” to shift the culture from manual to digital, but administrative tasks still require manual forms currently.
- **Governance:** All proposed systems must pass the IT Steering Committee via a Feasibility Study Report and approved based on necessity and department expertise.
- **Operations:** Departments operate in horizontal silos. The LMS and TCIS are currently run as separate business entities with disconnected workflow.

2. Application Architecture

- **Core system:** Legacy TCIS based on client-server technology.
- **Accessibility:** Restricted on campus Local Area Network (LAN). Staff cannot access the system while outstation without a Virtual Private Network (VPN) , this leads to bad user experience.
- **Integration:** Applications are fragmented. For example the LMS is a separate application from the main student database, requiring Application Programming Interface (API) to bridge the gap.

3. Data Architecture

- **Storage:** Data is distributed and decentralized. For example the student marks and physical asset record are separate, siloed databases.
- **Synchronization:** The data transfer relies on API calls or need manual entry , rather than real-time synchronization.
- **Performance:** Current database structure struggles with high concurrency or usage, causing up to 5- minute lags during peak usage.

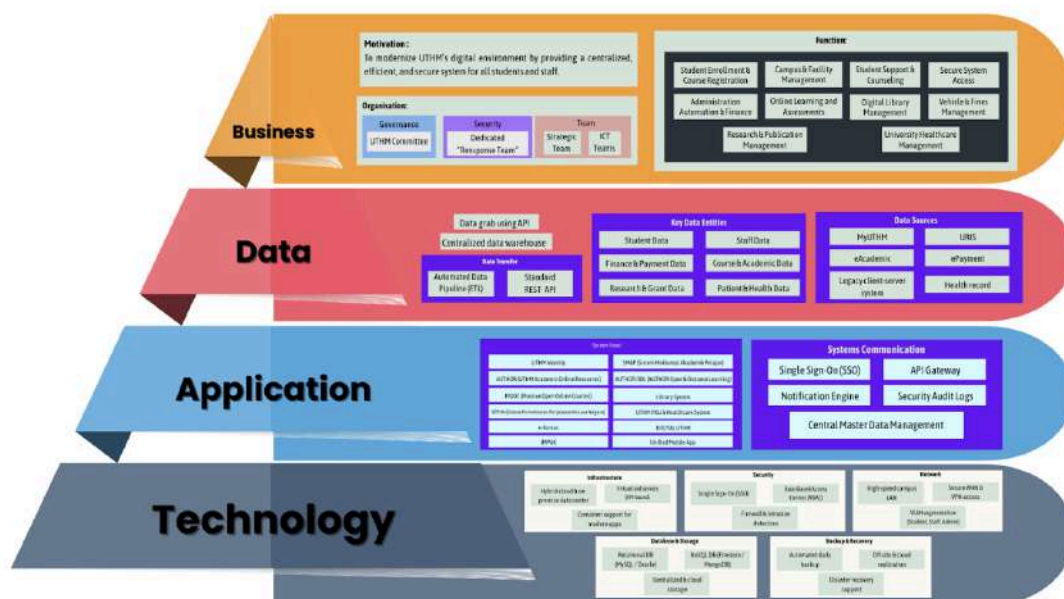
4. Technology Architecture

- **Facility:** Tier 3 Data Center located on the upper floor (flood prevention).
- **Server Hosting:** Uses Hot Aisle/Cold Aisle Containment for thermal efficiency.
- **Power:** High availability via Dual Power Sources (TN A/TN B), backed by UPS and Generator Sets.
- **Security:** Perimeter protection using 2 Firewalls (Europe/China brands) and a WAF for portal defense.

4.7.3 Enterprise Architecture To-Be

Figure 4.2 illustrates the Enterprise Architecture To-Be for UTHM. It represents the future system structure of the university. This architecture helps improve system efficiency and supports UTHM's future digital development.

Figure 4.2 : UTHM's To-Be Enterprise Architecture



4.7.4 System Design To-Be

This section explains the Enterprise Architecture To-Be as shown in Figure 4.2. The framework is organized into four main layers. Each layer serves a specific purpose.

1. Business Architecture

This layer, shown in green, describes what the university does and how it operates:

- Motivation: To develop a centralized, secure and efficient digital environment accessible to students and staff on any device at any time.
- Organization: Illustrates UTHM's departmental hierarchy.
- Functions: The architecture supports ten core business functions.
- Student Enrollment & Course Registration: Manages student admissions, course selection, and registration processes in a streamlined manner.
- Administration Automation and Finances: Manages administrative tasks, payroll, payment processing and allocation of university resources.
- Online Learning and Assessment: Support student learning and coursework using online platform.
- Research and Publication Management: Supports research project management, grant applications and innovation initiatives.
- Campus and Facility Management: Coordinates maintenance, space allocation, and utilization of campus facilities.
- Student Support and Counselling: Manages student welfare services, counselling sessions, and academic support programs.
- Secure System Access: Authorized access to systems through identity management and authentication mechanisms.
- University Healthcare Management: Supports the management of healthcare services and medical records within the university.
- Digital Library Management: Provides access to digital academic resources, journals and library services.
- Vehicle and Fines Management: Manages vehicle registration, parking systems and fine enforcement processes.

2. Information System Architecture

Divided into Data Architecture and Application Architecture:

- Data Architecture: Includes a centralized data warehouse that consolidates data from multiple systems. It supports dashboards for data visualization, automated data pipeline and data protection mechanisms. REST APIs are used to enable secure and standardized data exchange between systems.
- Application Architecture: The Application Architecture consists of multiple application that support academic, administrative, and operational activities at UTHM.
- Student & Academic Systems: UTHM Identity, SMPA, MyUTHM Academic, Open and Distance Learning (ODL) platform, Massive Open Online Courses (MOOCs), Author ID, and online academic resources.
- Administrative & Support Systems: Counselling Management System (CNS), SPLPN, UTHM Healthcare System, e-Kursus, DIGITAL UTHM, IMPAK, and Vehicle and Facilities Management systems.
- Library & Access Systems: Library System and Unified Mobile Application. Provide centralized access to academic and student services.
- Integration: Connects all systems through Single Sign-On (SSO), API Gateway, notification engine, security audit logs, and centralized master data management

3. Technology Architecture

Defines the infrastructure supporting all systems and is organized into seven domains:

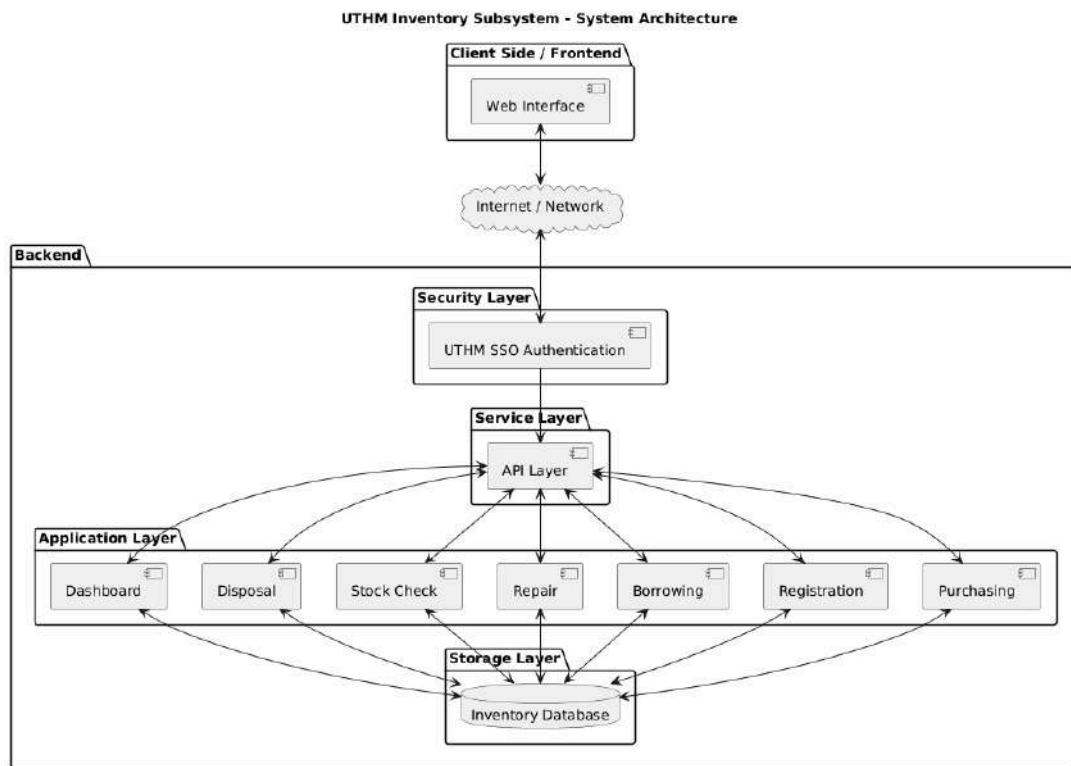
- Infrastructure: Hybrid cloud access and on-premise data center with virtualization support to ensure scalability, flexibility and efficient resource utilization.
- Network: High-speed backbone network, secure Wan and VPN access to support secure and advanced connectivity across the campus.
- Platform: Linux and Windows servers and web and mobile applications to support the deployment and execution of modern applications..

- Database: Relational (MySQL/Oracle), NoSQL (Firestore/MongoDB) and cloud storage for high availability.
- Application Hosting: Web servers such as Apache and Nginx, cloud-based hosting services, and API Gateway to host and integrate university applications.
- Security: Single Sign-On (SSO) , secure internal communications, firewalls, and intrusion detection to protect applications, data, and network resources.
- Backup: Automated backups, offsite/cloud replication, and disaster-recovery support data protection and business continuity.

4.7.5 System Architecture

Figure 4.3 below illustrates the system architecture of the UTHM Inventory Subsystem. Users access the system through the Web Interface, and all requests are transmitted via the Internet. Users are first authenticated using UTHM SSO before any system functionality can be accessed. Once authenticated, requests are processed by the API Layer, which routes them to the appropriate application modules. These modules execute the required business logic and interact with the Inventory Database to retrieve or update data. The processed results are then returned through the API Layer to the frontend for display.

Figure 4.3 : UTHM Inventory System Architecture



4.7.6 System Architecture Components

The UTHM Inventory Subsystem is designed using a cloud architecture. As shown in Figure 4.3, it divides the system into presentation which is the frontend, security, service, application and data layers. This design has a clear separation of responsibilities, maintainability and secure access to inventory data. The following explains the components in each layer and their functions.

1. Presentation Layer (Client Side / Frontend)

- Component: Web Interface
- Function: The presentation layer is the user-facing part of the system, where staff and students interact with the inventory subsystem. It handles input forms, dashboards and reports. It also presents all the data and insights in an intuitive and readable format. The frontend communicates with the backend over the network but does not perform any business logic or data storage operations.

2. Transport Layer (Internet / Network)

- Component: Internet or Network Cloud
- Function: This layer provides the connectivity between the frontend and backend. It enables secure transmission of requests and responses and allows users to access the system remotely while protecting data and confidentiality during transit.

3. Security Layer

- Component: UTHM SSO Authentication
- Function: Security is the first point of access for any system request. The UTHM SSO verifies user credentials and enforces access control to ensure that only authorised personnel can perform operations such as purchasing, borrowing or stock checking. Authentication occurs before any business logic is executed to provide a secure foundation for the system.

4. Service Layer (API Layer)

- Component: API Layer
- Function: The service layer acts as a gateway between the frontend and the application modules. It receives authenticated requests from the user interface, validates requests and routes them to the appropriate business logic modules and returns processed results to the frontend. This separation ensures that the frontend does not directly access backend modules or databases. This way, security risks are reduced and modularity is achieved.

5. Application Layer (Business Logic Layer)

- Components: Purchasing, Registration, Borrowing, Repair, Stock Check, Disposal, Dashboard
- Function: The application layer contains the core business logic of the inventory system. It consists of eight functions:
 1. Purchasing: To add new inventory items.
 2. Registration: To manage registration of users or items.
 3. Borrowing: To track check-outs and returns.
 4. Repair: To handle repair requests for damaged items.

5. Stock Check: To provide real-time inventory status and reports.
6. Disposal: To manage retirement of obsolete items.
7. Dashboard: To summarize key metrics for management.

All modules interact with the data layer to read or update inventory records that are being made by the administrator, lecturers or students of UTHM.

6. Data Layer (Storage Layer)

- Component: Inventory Database
- Function: The data layer stores all persistent system data including inventory records, user information and transaction history. It establishes data consistency, integrity and availability for all application modules.

4.7.7 Project Design

This section consists of Use Case Diagram, Activity Diagram and Sequence Diagram for Stock Checking under UTHM's Inventory Management System.

Use Case Diagram

Figure 4.4 : *Use Case Diagram for Stock Checking In Inventory*

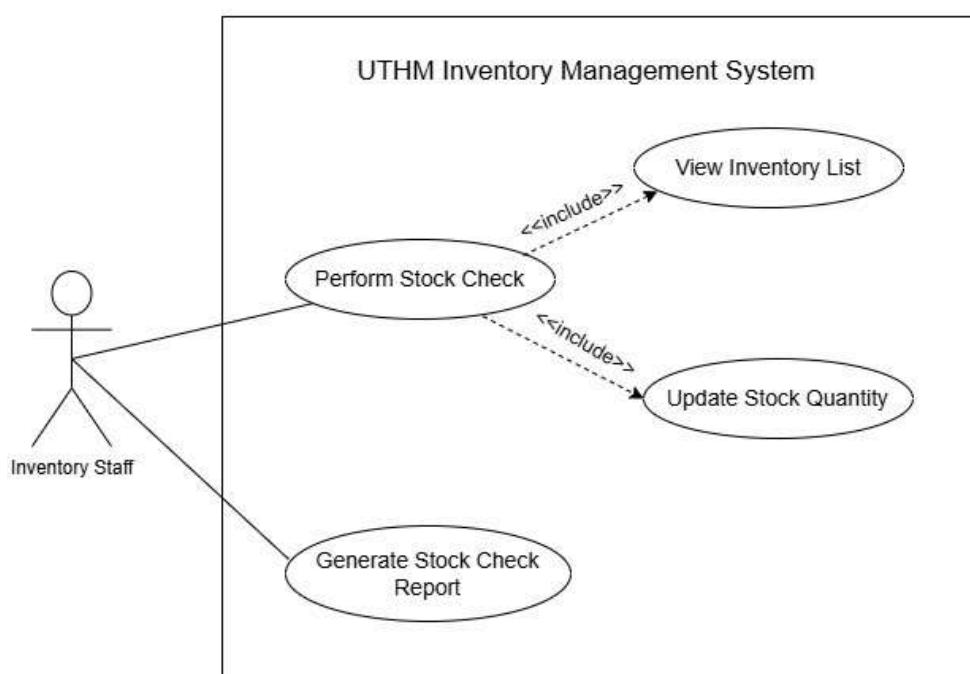


Figure 4.4 shows the use case diagram where the inventory staff access the UTHM Inventory Management System to view registered inventory items and perform physical stock checks. During the process, verified quantities are updated in the system. Apart from that, inventory staff can also generate stock check reports for auditing purposes.

Activity Diagram

Figure 4.5 : Activity Diagram for Stock Checking In Inventory

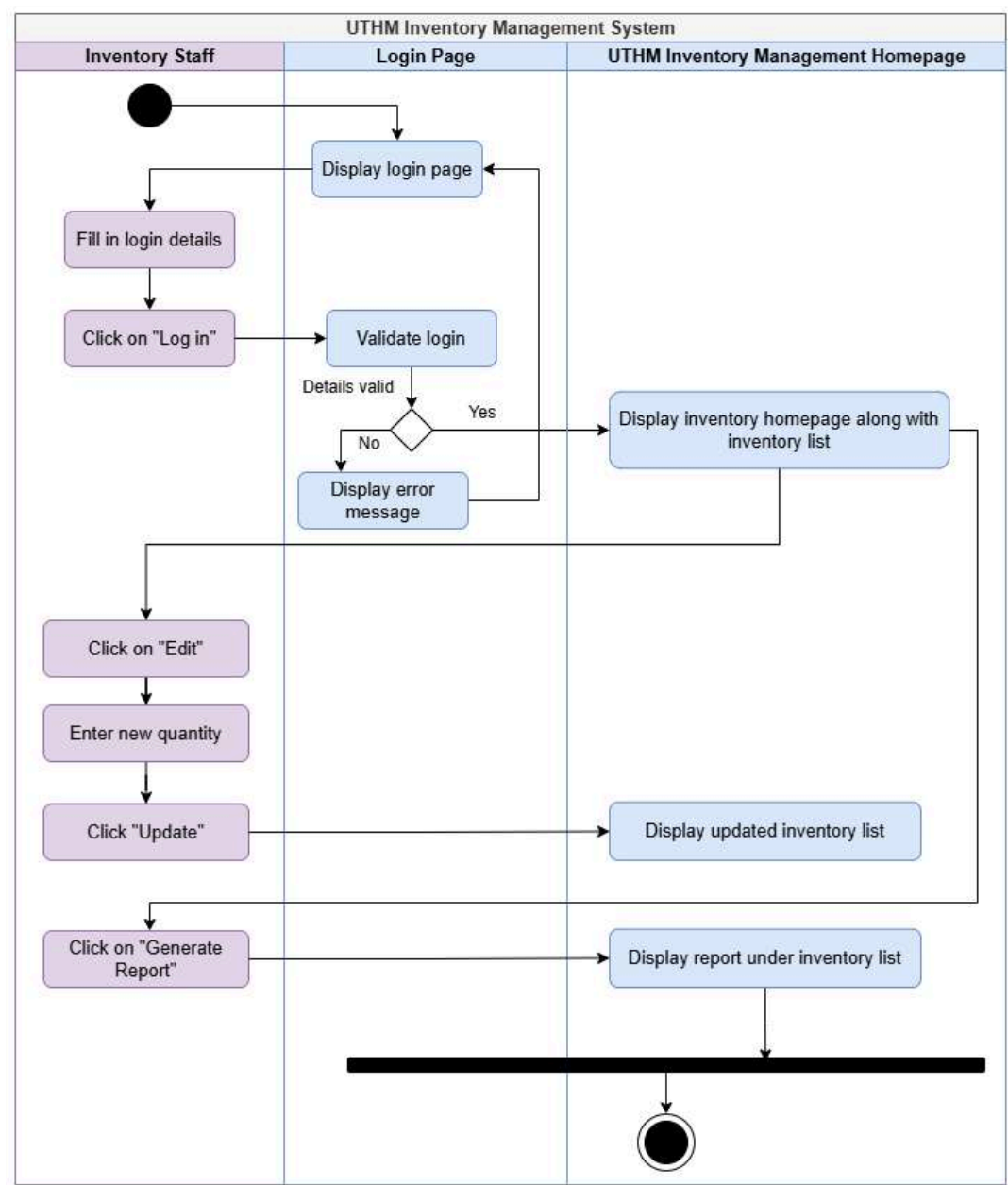
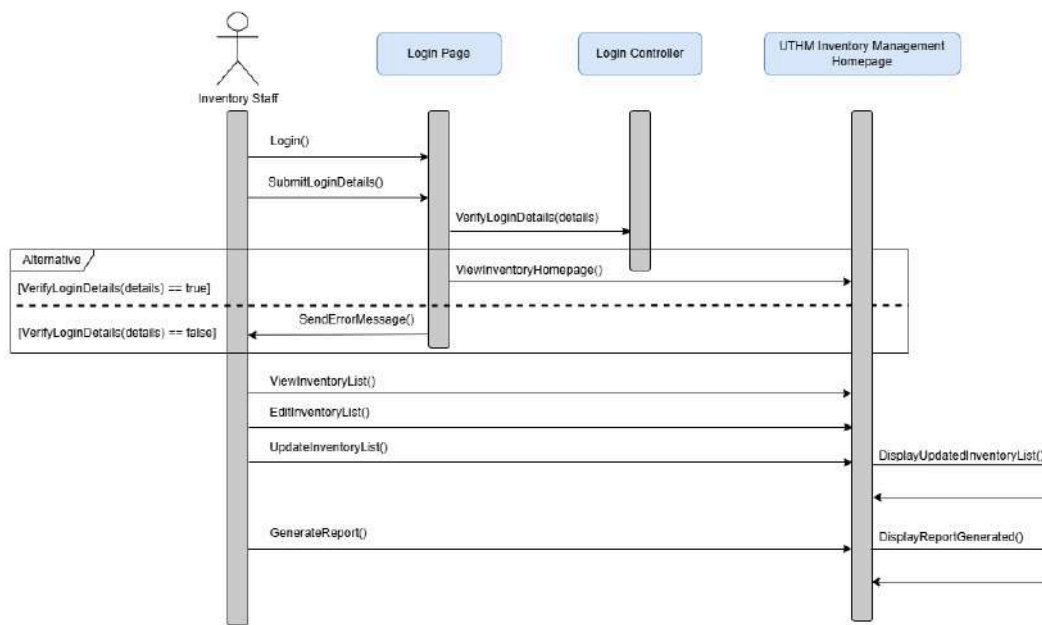


Figure 4.6 : Sequence Diagram for Stock Checking In Inventory



When the staff clicks the link for UTHM Inventory Homepage, it directs to the login page first. Staff are required to fill in their login details. If their login details are wrong, it displays an error message and redirects to the login page again. If the staff credentials are correct, staff will see the homepage of UTHM Inventory System where it displays the inventory list. If the staff wishes to update the quantity of the items, they are required to click on the “Edit” button first. Then, they can enter the updated quantity of the item and click the “Update” button to ensure it is updated in the system. If the staff wishes to generate the report, they should click on the “Generate Report” button under the inventory list. The report generated will then be displayed on the homepage.

4.7.8 Database Design

This section consists of Class Diagram, ERD and Star Schema for Stock Checking under UTHM’s Inventory Management System.

Class Diagram

Figure 4.7 : Class Diagram for Stock Checking In Inventory

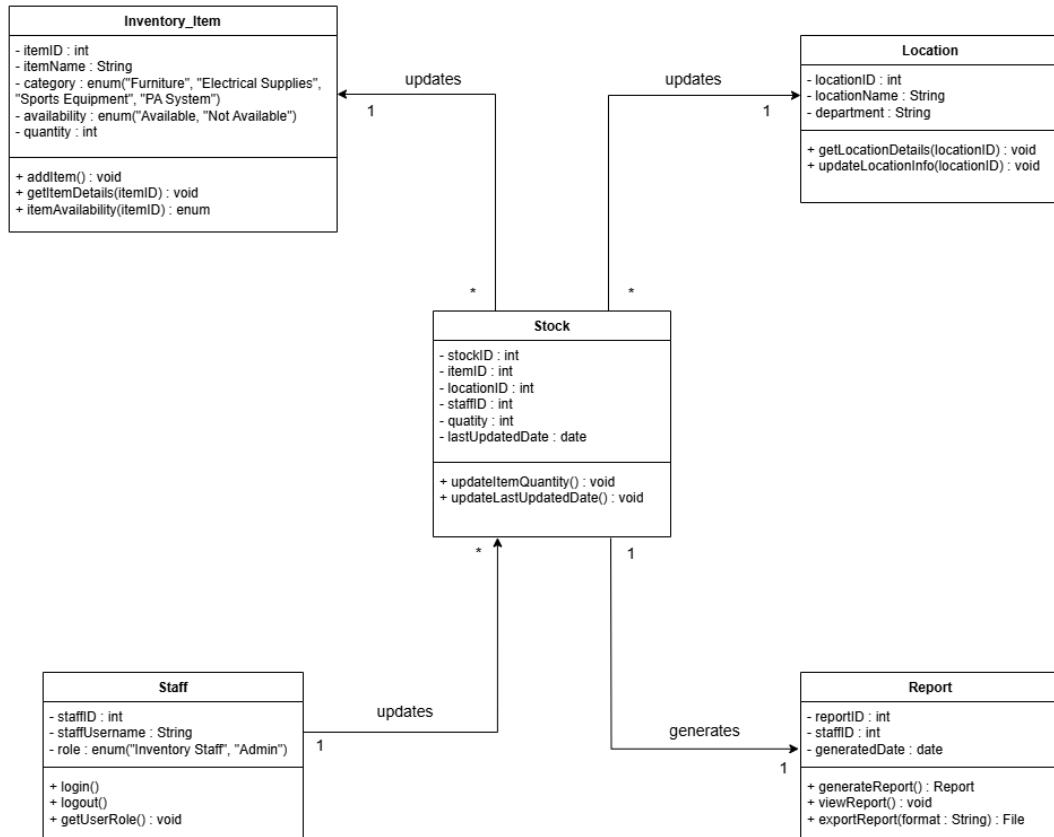


Figure 4.7 shows the structural design of UTHM's Inventory Management System. There are five tables named `Inventory_Item`, `Location`, `Stock`, `Staff` and `Report`.

The **Inventory_Item** class represents the master data of inventory items. It stores inventory item details and provides methods to add new item information, retrieve item information and item availability. This class ensures consistency of item data across the system.

The **Location** class models the physical storage locations within UTHM, including departments. It maintains location-related information and supports methods to retrieve and update location details. This allows the system to accurately track where inventory items are stored.

The **Staff** class represents system users such as inventory staff and administrators. It manages user authentication and authorization through methods such as login and

role identification. This class supports accountability by linking stock check and report generation activities to specific users.

The **Stock** class is the core operational component of the system. It handles inventory stock checking processes by recording item quantities at a specific location. Methods in this class support updating item quantity and updating the latest stock check date, ensuring accurate and up-to-date inventory data.

The **Report** class is responsible for managing inventory stock check reports. It generates reports based on verified stock data, allows users to view report details, and supports exporting reports. Separating report functionality into its own class improves modularity and aligns with best practices in system design.

Entity Relationship Diagram (ERD)

Figure 4.8 : *Entity Relationship Diagram for Stock Checking In Inventory*

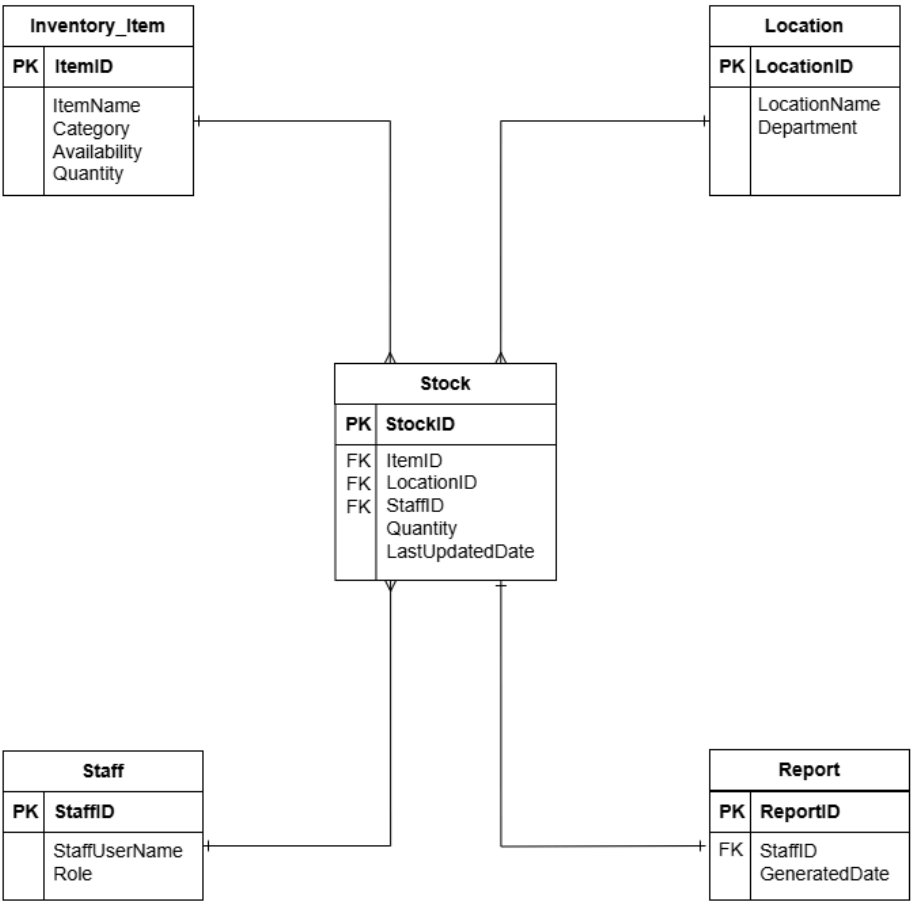
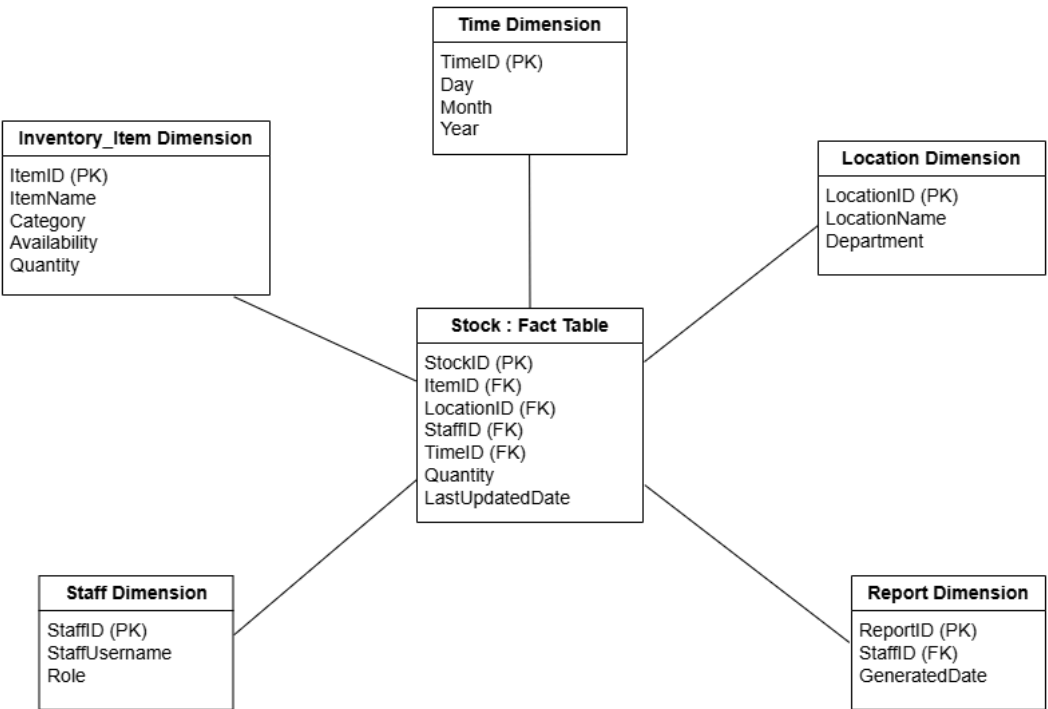


Figure 4.8 shows the ERD that illustrates the database structure of the UTHM Inventory Management System. It shows how inventory items, locations, users, stock records, and reports are stored and related to support inventory management activities. The InventoryItem entity stores master data of items, while the Location entity represents physical storage locations within UTHM. The Staff entity records system users responsible for stock checks and report generation. The Stock entity acts as the central transactional table, linking items, locations, and users, and storing system quantities. The Report entity manages inventory reports generated from stock data and records report details. Overall, the ERD ensures data integrity through clearly defined one-to-many relationships and one-to-one relationships and supports efficient stock verification and reporting for UTHM.

Star Schema

Figure 4.9 : Star Schema of Stock Checking In Inventory



The star schema represents the analytical data model for the UTHM Inventory Management System. The Stock : Fact Table stores quantitative measures such as

system quantity and last updated date from stock checks. Surrounding dimension tables provide descriptive context related to inventory items, locations, users, reporting details, and dates.

4.7.9 Interface Design

Figure 4.10 : Interface Design of UTHM Inventory Management Homepage for Inventory Staff

Item ID	Item Name	Category	Availability	Quantity
FN001	Office Desk	Furniture	Available	15
FN002	Bookshelf	Furniture	Available	10
ES001	Extension Cord	Electrical Supplies	Available	25
ES002	Power Drill	Electrical Supplies	Not Available	0
SP001	Basketball	Sports Equipment	Not Available	5
SP002	Treadmill	Sports Equipment	Available	3
PA001	Microphone	PA System	Available	8
PA002	Loudspeaker	PA System	Available	4

Figure 4.10 shows the UTHM Inventory Homepage interface built using SAP. It displays a list of inventory items with Item ID, Item Name, Category, Availability Status, and Quantity. The categories shown include furniture, electrical supplies, sports equipment, and PA system items. The availability status is not dependent on quantity, allowing items such as sports equipment to be marked as Not Available even when stock exists. For example, when items are fully booked, it is also considered as not available. The “Edit” button below the list allows authorized users to modify inventory information and the “Generate Report” button enables staff to generate inventory reports for monitoring and auditing purposes.

Figure 4.11 : Interface Design of UTHM Inventory Management Homepage for Admin

UTHM Inventory Management System

Edit Inventory Quantities

Item ID	Item Name	Category	Availability	Quantity
FN001	Office Desk	Furniture	Available	15
FN002	Bookshelf	Furniture	Available	10
ES001	Extension Cord	Electrical Supplies	Available	25
ES002	Power Drill	Electrical Supplies	Not Available	0
SP001	Basketball	Sports Equipment	Not Available	5
SP002	Treadmill	Sports Equipment	Available	3
PA001	Microphone	PA System	Available	8
PA001	Loudspeaker	PA System	Available	8
PA002	Loudspeaker	PA System	Available	4

+ Add Item **Update**

Figure 4.11 shows the interface of Edit Inventory for admins only who can directly update item quantities. The inventory table layout remains consistent but the quantity column is editable, allowing adjustments based on stock checks, returns, or new items received. The “Add Item” button enables the admin to add a new item into the inventory list whereas the “Update” button allows staff to confirm and save all quantity changes in the system. This design helps maintain controlled access, ensuring that only authorized users can modify inventory records within the system.

4.8 Chapter Summary

This chapter presents the overall analysis and design of the proposed system for the UTHM case study. It begins with an introduction to the organization and its structure, followed by an analysis of the current system to identify existing limitations and challenges. A comparison between the existing and proposed systems is then discussed to highlight the improvements introduced by the new solution. The chapter also explains the techniques used for system requirements gathering and outlines the

identified system requirements. Subsequently, the system design is presented, covering both the AS-IS and To-Be enterprise architectures, system designs, and overall system architecture. The chapter concludes with detailed discussions on project design, database design, and interface design to demonstrate how the proposed system supports UTHM's operational and digital transformation needs.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 Introduction

This chapter explains the development of the UTHM Inventory Management System using SAP Build Apps. It provides a step-by-step overview of how the system was designed. The creation of the stock checking module is emphasized in order to demonstrate how inventory data is retrieved, displayed, and managed within the system. This chapter's overall goal is to provide a clear understanding of the development process and how SAP Build Apps supports rapid application development for enterprise systems.

5.2 System Development

This report focuses on describing the development and functionality of an inventory system built using SAP Build Apps. The system is designed to support inventory management processes through seven core features, namely the dashboard, registration, purchasing, borrowing, repair, stock checking, and disposal modules. Each feature plays a specific role in ensuring efficient tracking and control of inventory items. This section specifically discusses the stock checking feature, which enables administrators and staff to monitor item availability and quantities accurately, supporting effective decision-making and operational efficiency.

5.3 Create Database

Once signing in into SAP Build Apps, a new database can be created under the Integration section as shown in Figure 5.1. A new database named InventoryData was created for this project. The database consists of seven fields which are itemID which is the primary key, itemName, category, location, quantity, itemsetID which is

the ID set by admins and will be displayed in the system later and lastly, isBookedOut. itemID, itemsetID, itemName, category and location variables are text data type, quantity variable is number data type and isBookedOut variable is boolean data type. Figure 5.2 shows the fields in the InventoryData database.

Figure 5.1 : Integration Section of SAP Build Apps

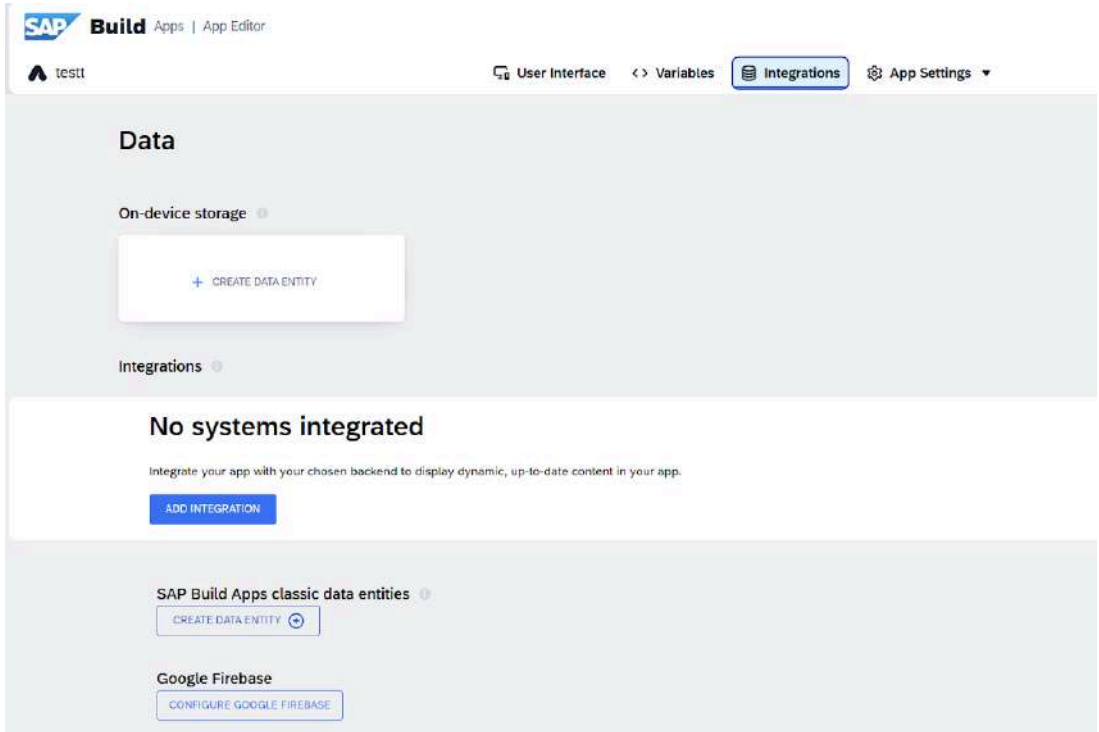
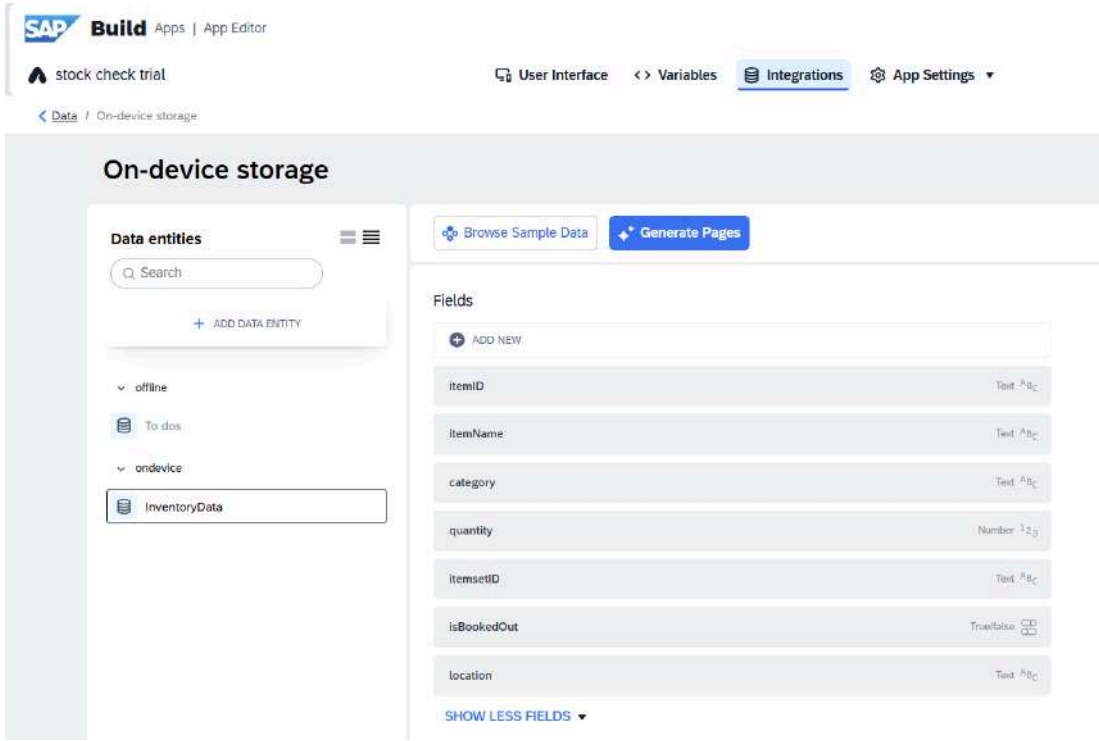


Figure 5.2 : Fields in InventoryData Database



5.4 Coding Of The System

After creating the database, an admin role is created to differentiate normal inventory staffs and administrative staff. The difference between inventory staff and administrative staff is that administrative staff can add new items into the inventory list. To create the admin role, a new application variable named `isAdmin`. The value type of the variable is set to true or false. Then, a data variable named `InventoryData1` is added which reads the data from database `InventoryData`. Figure 5.3 and 5.4 shows that these steps have been done.

Figure 5.3 : Created `isAdmin` Role

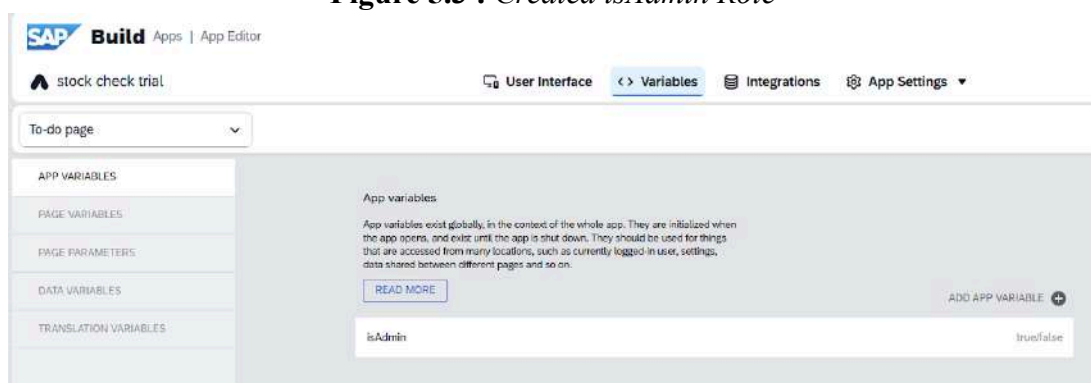
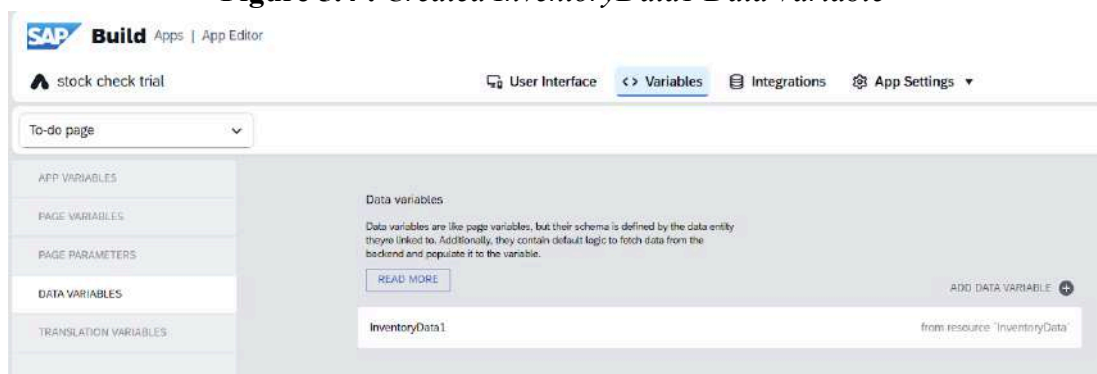


Figure 5.4 : Created `InventoryData1` Data Variable



Next, the interface where the inventory list will be displayed has to be built. A container with six fields named, ID, Item, Category, Location, Quantity and Availability has been added as the header of the interface. Then, another container with six text fields was added to represent displaying the inventory list. Each text field was bound to the data variable `InventoryData1`, thus displaying the data in the database in the user interface. The availability of the items does not only depend on

the quantity of the items but also whether they are booked out. For example, if the university has five speakers and all five of them are booked out, then the quantity will be displayed as ‘5’ and the availability will be displayed as ‘Not Available’ as none of them are ready to use. Hence, to make sure this logic works, the text field for isBookedOut was bound to a formula. The formula as is the following :

IF(repeated.current.quantity == 0 || repeated.current.isBookedOut == true, "Not Available", "Available")

An “Edit Items” button was added to allow administrative staff to add new items into the inventory list. To make sure this button was only visible to admins, the logic of the button was set as *appVars.isAdmin*. This statement means that the button will only appear if isAdmin is true, otherwise the button will not be displayed. Finally, a “Generate Report” button was added. Figure 5.5 shows the overall features of the system whereas Figure 5.6 shows the interface with sample data.

Figure 5.5 : Components Of Inventory System Homepage

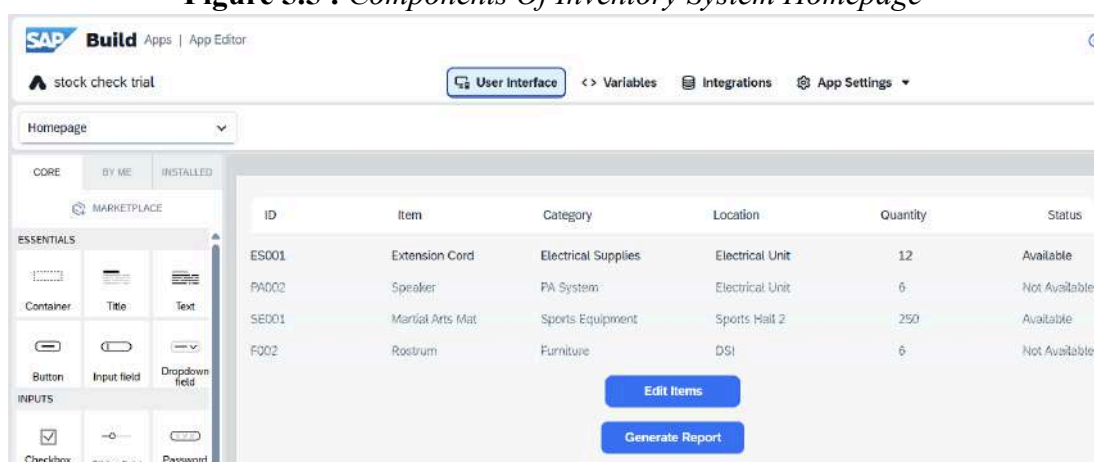
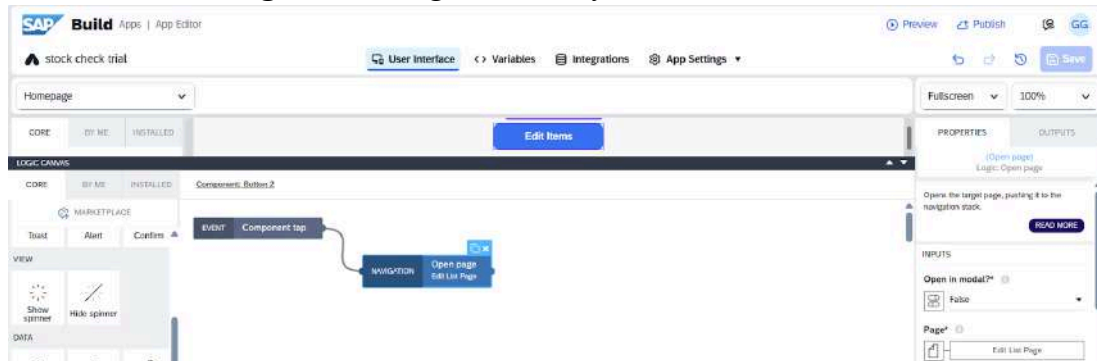


Figure 5.6 : Inventory System Homepage Interface With Sample Data

ID	Item	Category	Location	Quantity	Status
ES002	Projectors	Electrical Supplies	Electrical Unit	8	Available
F001	VIP Chairs	Furniture	DSI	50	Available
F002	Rostrum	Furniture	DSI	6	Not Available
SE001	Martial Arts Mat	Sports Equipment	Sports Hall 2	250	Available
PA002	Speaker	PA System	Electrical Unit	6	Not Available
ES001	Extension Cord	Electrical Supplies	Electrical Unit	12	Available
PA001	Microphone	PA System	Electrical Unit	15	Available

Upon clicking on the “Edit Items” button, the system will redirect the administrative staff to a new page where they can edit the location, quantity and availability of the items. In the Logic Canvas of the “Edit Items” button, a flow is created from “Component Tap” to “Navigation” which opens up to a new page named “Edit List Page”. Figure 5.7 shows the Logic Canvas of “Edit Items” button.

Figure 5.7 : Logic Canvas of “Edit Items” Button



In the “Edit List Page”, there are six fields named Item ID, Item Name, Category, Location, Quantity and Availability and lastly has an “Update” button in each row. The item ID, item name and category are read only, whereas the location, quantity and the availability can be modified. Once admin is done editing for the particular item, they have to click on the “Update” button to save the changes. If the update was successful, the system will redirect to the homepage. If the admin wants to add a new item to the list, they can click on the “Add Item” button at the bottom of the page. The system will redirect to a new page named “Add Page”. Figure 5.8 shows the interface for “Edit Page”.

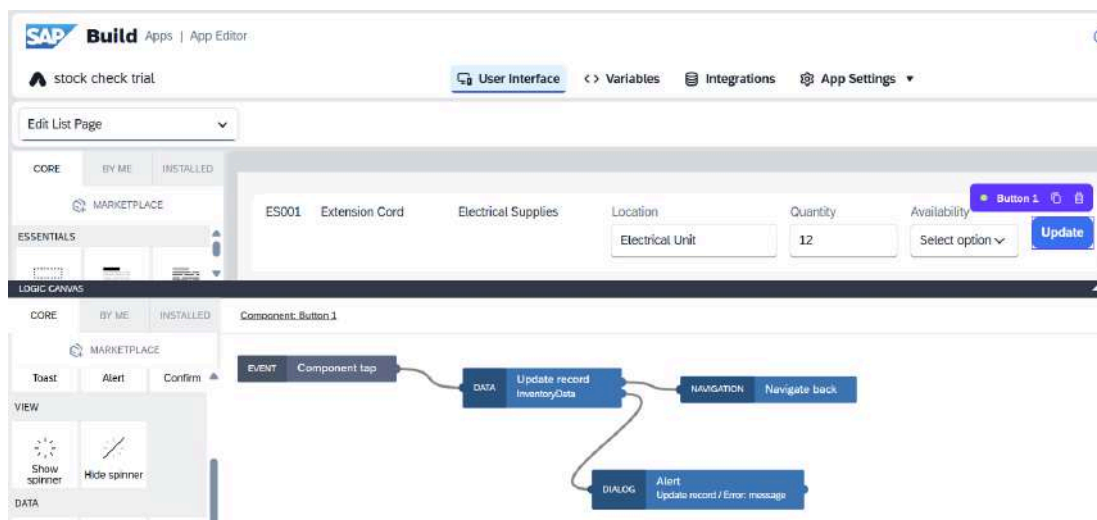
Figure 5.8 : Interface For Edit List Page

Item ID	Item Name	Category	Location	Quantity	Availability	Action
ES002	Projectors	Electrical Supplies	Electrical Unit	9	Select option	Update
F001	VIP Chairs	Furniture	DSI	50	Select option	Update
F002	Rostrum	Furniture	DSI	6	Select option	Update
SE001	Martial Arts Mat	Sports Equipment	Sports Hall 2	250	Select option	Update
PA002	Speaker	PA System	Electrical Unit	6	Select option	Update
ES001	Extension Cord	Electrical Supplies	Electrical Unit	12	Select option	Update
PA001	Microphone	PA System	Electrical Unit	15	Select option	Update

[Add New Item](#)

As for the Logic Canvas for the “Update” button, the flow is created from “Component Tap” to “Update Record” which updates the existing record in the database, to “Navigation” so that the system navigates back to the homepage. An “Alert” is also dragged from the “Update Record” tab where it will display an error message if any field doesn’t comprise the data format. Figure 5.9 shows the Logic Canvas of the “Update” button.

Figure 5.9 : Logic Canvas of “Update” button



If the admin clicks on the “Add Item” button, the system will redirect to the “Add Page”. In the “Add Page”, there are six fields named Item ID, Item Name, Category,

Location, Quantity and Availability. Once admin is done filling up the information for the new item, they have to click on the “Save Item” button to save the changes. If the update was successful, the system will redirect to the homepage. Figure 5.10 shows the interface for “Add Page”.

Figure 5.10 : Interface For Add Page

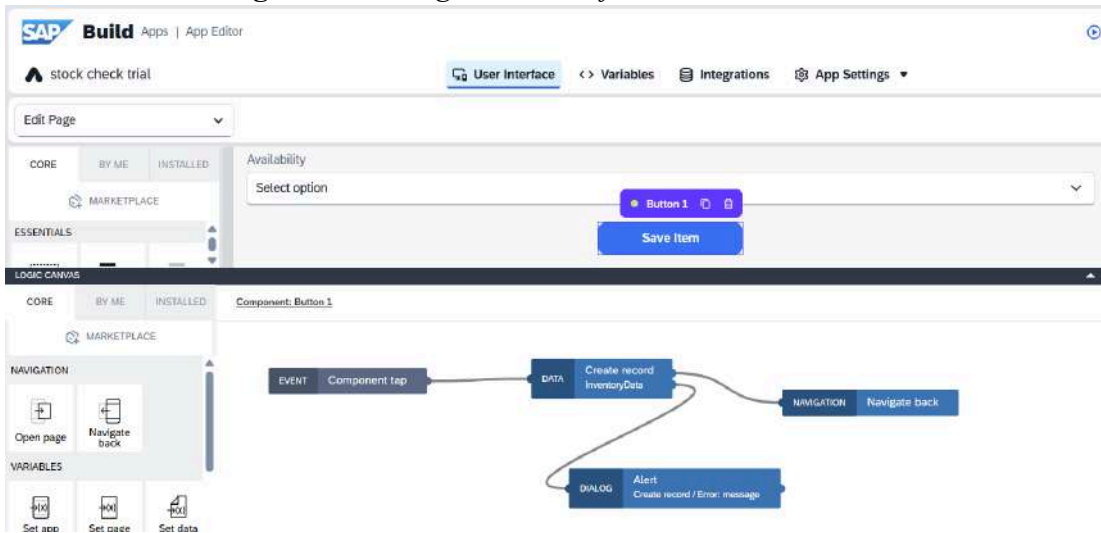
The screenshot displays a form titled "To-do page" with the following fields:

- Item ID: Type here...
- Item Name: Type here...
- Category: Type here...
- Location: Type here...
- Quantity: Type here...
- Availability: Select option (dropdown menu)

A blue "Save Item" button is located at the bottom right of the form.

As for the Logic Canvas for the “Save Item” button, the flow is created from “Component Tap” to “Create Record” which creates the new record in the database, to “Navigation” so that the system navigates back to the homepage. An “Alert” is also dragged from the “Create Record” tab where it will display an error message if any field doesn’t comprise the data format. Figure 5.11 shows the Logic Canvas of the “Save Item” button.

Figure 5.11 : Logic Canvas of “Save Item” Button

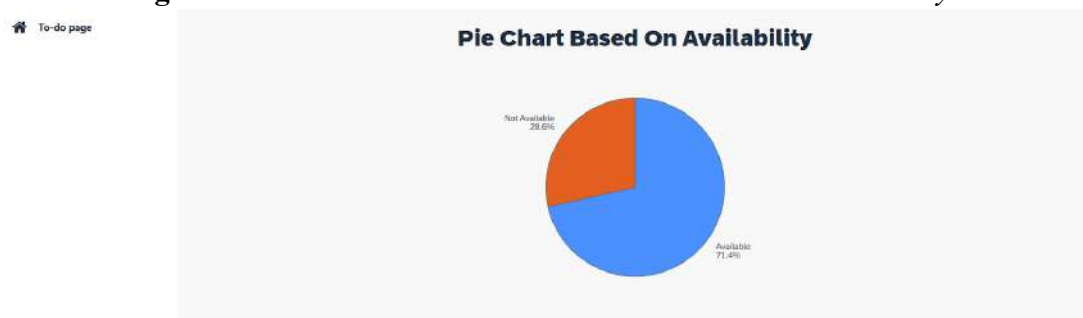


As for the “Generate Report” button, the system will also redirect to a new page which displays a pie chart of the availability of the items. The pie chart feature was installed and used. The formula used to create the pie chart is as the following :

```
[ { label: "Available", value: COUNT(SELECT(data.InventoryData1,
!item.isBookedOut && item.quantity > 0)) },
{ label: "Not Available", value: COUNT(SELECT(data.InventoryData1,
item.isBookedOut || item.quantity == 0)) } ]
```

The label is the display name of the variables in the pie cart being “Available” and “Not Available”.

Figure 5.12 : Pie Chart Generated Based On Items' Availability



CHAPTER 6

CONCLUSION

6.1 Introduction

This final chapter provides a reflective summary of the UTHM Inventory System project. It outlines the primary achievements of the development process, acknowledges the limitations encountered during the implementation in the SAP Build Apps environment, and proposes strategic suggestions for future enhancements. The conclusion reinforces how the system fulfills the original objective of digitizing the manual inventory and its processes at the university.

6.2 System Contribution/Achievement

The UTHM Inventory System has successfully introduced several key improvements to the management of university assets:

- **Centralized Data Management:** The transition from manual records to a structured relational database (On-Device Storage) ensures that user, item, and transaction data are synchronized and easily retrievable.
- **Automation of Approval Workflows:** By implementing sequential logic, the system automatically updates item availability statuses upon clerk approval, reducing administrative workload and preventing data entry errors.
- **Enhanced User Transparency:** Borrowers can now view real-time item availability and report damages directly through the mobile interface, fostering better accountability for university property.
- **Real-time Damage Reporting:** The integration of a dedicated damage reporting module allows the maintenance team to identify and isolate faulty equipment immediately.

- **Better Decision Making and Planning:** The management can make an accurate decision regarding item procurement, replacement strategies, allocation of campus resources and also long term infrastructure planning.
- **Increased Security and Reliability:** Supported by secure network infrastructure, backup systems, and disaster recovery mechanisms, the subsystem ensures that critical inventory data remains protected, available, and recoverable at all times.

6.3 System Constraint

Despite the successful deployment of the prototype, several constraints were identified:

- **Offline Data Dependency:** As the system currently utilizes On-Device Storage, data is localized to the device. Real-time synchronization across multiple different clerk devices would require an external cloud database integration.
- **Cost Implementation:** By implementing SAP Business Technology Platform (BTP) environment requires a significant amount of investment and long-term commitment from the IT steering committee.
- **Technical Skill Requirement:** The system requires personnel with adequate technical knowledge to maintain, configure and troubleshoot the application. Limited expertise in SAP BTP and related technologies may slow down system enhancements and issue resolution.
- **Limited System Integration:** The current prototype operates as a standalone system and has limited integration with existing organizational systems. This may cause data error and reduce efficiency until proper system integration is implemented.

6.4 Future Suggestion

To further evolve the UTHM Inventory System, the following enhancements are recommended:

- **Cloud Backend Integration:** Transitioning the database to a cloud-based service that is also available in SAP BTP to allow real-time data sharing between students and multiple administrative departments.
- **QR Code/Barcode Integration:** Implementing a scanning feature using the device camera to allow users to scan an asset tag, which would automatically pull up the item's details and all the information users need to know.
- **Deep Integration with Financial Systems:** The system should further evolve to integrate directly with the Financial System via API. This would allow the automated calculation of asset depreciation and block the student account in LMS if they fail to return borrowed equipment on time.
- **Internet of Things (IoT) & RFID Tracking:** To automate the stock checking process, future iterations might upgrade from passive QR codes to active RFID tags. By installing the IoT sensor in laboratories, the system could auto detect the position of the high value asset , and no more manual scan of each asset by staff.
- **Predictive Analytics and Smart Dashboard Enhancement:** The system could incorporate predictive analytics powered by SAP analytics tools to forecast asset usage, maintenance cycles, and replacement needs. By analysing historical data from borrowing, repair, and disposal modules, the dashboard can provide insights such as high-demand items, frequently damaged assets, and upcoming procurement requirements. This would support proactive decision-making by administrators and reduce unexpected asset shortages.

6.5 Summary

In summary, the UTHM Inventory System project has successfully demonstrated the efficacy of using low-code development to solve campus administrative challenges. By streamlining the processes and digitizing inventory records, the system provides a robust framework for asset management. While certain technical constraints exist regarding data synchronization, the proposed future suggestions provide a clear roadmap for scaling the application into a comprehensive university-wide solution.

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